

HelixVPN MVP — Source-Coverage Ledger (§11.4.118 proof-of- completeness)

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Revision: 1 **Last modified:** 2026-06-25T00:00:00Z **Status:** active **Authority:** Constitution §11.4.118 (discovery-pressure / enumerated-coverage), §11.4.6 (no-guessing — gaps surfaced, never hidden) **Scope:** Enumerated proof that every one of the 16 source research documents under docs/research/mvp/ (11 LLM analyses + 5 04_VPN_CLD refined docs) was read and that its distinctive content is accounted for in the final/ specification set — OR is named as a gap. Evidence base: v09-research/_SYNTHESIS.md + citation-grep across all 13 produced final/*.md docs.

How to read this ledger

- **Source id** — the canonical id used in citations ([04_ARCH §3], [02_QWN], ...).
- **Absorbed into** — which final/ doc(s) carry its content (verified by source-id citation grep + topical reconciliation; the dominant landing site is **bolded**).
- **Unique contribution** — the distinctive idea(s) this source brought that the others did not (or brought first / most sharply).
- **Where it landed** — the concrete section/decision the contribution shaped.

final/ doc shorthand: **00** product-scope · **01** data-plane · **02** control-plane · **03** client-core+UI · **04** security/privacy/PKI · **05** repo-layout/tooling/ecosystem · **06** Phase-0 WBS · **07** Phase-1 WBS · **08** Phase-2 WBS · **09** Phase-3 WBS · **10** testing/QA · **11** deep-research appendix · SPEC SPECIFICATION.md (master index).

Coverage table — one row per source

Source id	Absorbed into (final/)	Unique contribution	Where it landed
00 (VPN_Initial_Res)	00 , 01, 02, 03, 04, 06 , 07 , 08, 09, 10, 11, SPEC	The founding constraint: 1 user → N joined private networks, no inbound port-forward (internal hosts dial outbound; reverse tunnel). Original problem statement + glossary all docs anchor to.	Product floor (00 §Problem), reverse-tunnel hub-and-spoke (01), threaded as the requirement source through every WBS.
01_DSK	00, 02, 11, SPEC	Go-core camp (D2 Camp B): Go client core reusing Hysteria2's Go impl — the simpler, single-language alternative to the Rust core. Lean tunnel-first MVP stance.	Decision D2 (client-core language) Camp B in 11-appendix + 00; lean-MVP argument feeding Phase-0/Phase-1 scoping.
02_QWN	00, 02, 03, 09, 10, 11, SPEC	Detailed control-plane modeling; Redis Streams-for-MVP → NATS-at-scale staging (D3 Camp A); event-bus + real-time push depth.	Decision D3 (event bus, Redis MVP) in 02 + 11; control-plane module decomposition (02).
03_ZAI	00 (1×), 11 (1×); DR/KMS content → 04 , 08	KMS-encrypted automated backups, Terraform disaster-recovery (RTO < 15 min) , GitOps config management, multi-arch CI/CD. Enterprise resilience/DR framing.	KMS backups + RTO in 04-security (RTO present); HA/multi-region + GitOps in 08-Phase-2. CI/CD content deliberately dropped per §11.4.156 (no active CI). See gap G1.
05_YBO	00 , 01, 02, 03, 04, 05, 07, 09, 10, 11, SPEC	The operator mandate : Go + Gin + PostgreSQL + Redis + rootless Podman stack floor; full platform list (iOS, Android, Aurora OS, HarmonyOS NEXT, Windows, Linux, macOS, Web); obfuscation mandatory "especially QUIC"; three apps.	Settled stack floor (00 §Stack), platform matrix (03 + 09), constitution-binding alignment (§11.4.76/.161). The most-cited mandate after 00/04_ARCH.

Source id	Absorbed into (final/)	Unique contribution	Where it landed
06_GRK	00 (1×), 11 (1×); protocol/hardening content → 01, 04, 08	sing-box as the universal transport-multiplexer framework; KMP (Kotlin Multiplatform) client-core alternative; detailed Podman-rootless firewall/netns hardening scripts; advanced Hysteria2 + Salamander configs.	KMP captured as a considered alternative in 03/11; Hysteria2/Salamander → D1 Camp B (01/08); rootless hardening → 04. sing-box framework itself not adopted (custom Rust helix-transport chosen instead) — see gap G2.
07_GMI	00, 03, 08, 09, 10, 11, SPEC	CGNAT 100.64/10 1:1-per-network answer to the IP-subnet-collision problem (D4 Camp B); Go-core leaning (D2 Camp B with DSK).	Decision D4 (subnet-collision) Camp B surfaced in 11 + IPAM discussion (02/08); one of only three sources that actually solved D4.
08_CPL	00 (2×), 11 (1×); connector mechanics → 01	Pragmatic minimal connector implementation : read routes from a file, enable forwarding, NAT LAN → tunnel — the smallest-viable network-side agent.	Subsumed into the Connector data-path + route-advertisement design in 01-data-plane ; no distinctive idea left unreflected (fully generalized).
09_GCT	00 (1×), 11 (1×); Go-only + SSE → 02, 03	Go-as-single-language stance; server-side kill-switch; SSE real-time channel; Fyne desktop UI toolkit; Postgres schema sketch (Russian-language source).	Go-only → D2 Camp B (11); SSE → real-time transport options (02 control-plane offers WS/SSE); server-side kill-switch reflected. Fyne explicitly overruled by Flutter and not recorded as a considered-then-rejected option — see gap G3.
10_KMI	00, 02, 03, 08, 09, 10, 11, SPEC	NATS-JetStream-from-start (D3 Camp B); CGNAT-per-	Decisions D3 Camp B + D4 Camp B + D7

Source id	Absorbed into (final/)	Unique contribution	Where it landed
11_MST	00, 01, 02, 03, 08, 09, 10, 11, SPEC	network (D4 Camp B, with GMI); full "Connectivity-OS" ambition (D7 Camp B).	Camp B in 11; scale/HA NATS path in 08-Phase-2; ambition framing in 00.
		Asymmetric per-leg transport (D6): Hysteria2/QUIC user gateway, WireGuard gateway networks — best-fit protocol per leg. Largest/most-detailed analysis.	Decision D6 (transport topology) in 11 + 01-data-plane (per-leg transport design); woven through Phase-2 transport set (08).
		The settled architecture spine : pitch, WireGuard-as-crypto-core + pluggable-obfuscation-beneath-WG, MASQUE/QUIC = WG-over-HTTP-3 (D1 Camp A = true Mullvad parity), IPv6 ULA /48 + 4via6 (D4 Camp A), repo layout, security invariants, no-logging-as-code.	The dominant citation across the whole set (50× in 00, 39× in 01, 32× in 04, 67× in 11). Anchors D1 Camp A, D4 Camp A, the repo layout (05), and the security model (04).
04_ARCH (CLD refined)	00, 01, 02, 03, 04, 05, 06, 08, 09, 10, 11, SPEC		
04_UI (CLD refined)	00, 03, 05, 09, 10, 11, SPEC	helix-core (Rust) + helix-ui (Flutter) shared-codebase strategy; helix_design system + signature components; three flavors from one tree (runHelixApp); Melos monorepo; per-platform TunnelPlatform shim.	The client architecture core of 03-client-core-and-ui ; design-system + platform-shim sections; Phase-3 platform builds (09).
04_P0 (CLD refined)	00, 01, 03, 06, 07, 10, 11, SPEC	Phase-0 spike with exit gates G1–G6 (G3 iOS NE memory ceiling = make-or-break; G4 Go-vs-Rust edge benchmark); surviving interfaces (Transport trait, helix-wg, orchestrator, FFI); Linux-netns + nftables-DPI + tc-netem test rig.	The spine of 06-Phase-0-WBS ; gate definitions also seed 10-testing; decision D5 (edge language) gate G4 in 01/11.

Source id	Absorbed into (final/)	Unique contribution	Where it landed
04_P1 (CLD refined)	00, 02 , 03, 04, 05, 07 , 08, 09, 10, 11, SPEC	Phase-1 MVP: Go modular monolith module list; Postgres+RLS data model (no connection/traffic tables, CI-lint-enforced); protobuf Coordinator + server-streaming WatchNetworkMap (p99 <1s); policy compiler (default-deny → AllowedIPs/nftables); 8-criterion MVP DoD.	The spine of 07-Phase-1-WBS + 02-control-plane ; the 8 acceptance criteria → 10-testing; no-logging schema lint → 04-security.
04_P2 (CLD refined)	00, 01, 04, 08 , 09, 11, SPEC	Phase-2 parity: full transport set (+Shadowsocks, UDP-over-TCP, hardened LWO, auto-ladder); DAITA via maybenot; direct P2P + NAT traversal + DERP-style relay; multi-hop; post-quantum (ML-KEM hybrid / Rosenpass); HA multi-region.	The spine of 08-Phase-2-WBS ; PQ + DAITA → 04-security; multi-region/HA → 08 + Patroni/NATS references.

Decision-coverage cross-check (the 8 D-decisions are all surfaced)

Decision	Camp A source(s)	Camp B source(s)	Surfaced in final/
D1 obfuscating transport	04_ARCH/04_P0 (MASQUE/QUIC)	plurality of 10 (Hysteria2+Salamander)	01, 08, 11
D2 client-core language	04_*/QWN/ZAI/CPL/KMI/MST (Rust)	DSK/GMI/GCT (Go)	03, 11
D3 event bus	04_P1/QWN/MST (Redis→NATS)	DSK/KMI (NATS-from-start)	02, 08, 11
D4 subnet collision	04_ARCH (ULA/4via6)	GMI/KMI (CGNAT 100.64/10)	02, 08, 11
D5 edge language	04_P0 (Rust)	(Go quic-go)	01, 06 (gate G4), 11

Decision	Camp A source(s)	Camp B source(s)	Surfaced in final/
D6 transport topology	single-protocol	11_MST (asymmetric per-leg)	01, 11
D7 MVP ambition	QWN/DSK/04_P0 (lean)	KMI/ZAI (Connectivity-OS)	00, 11
D8 licensing / positioning	04_ARCH §13 (source-available + commercial)	pure OSS	00, SPEC §9 (pre-Phase-3 decision)

All eight decisions (D1–D7 from SYNTHESIS §3 + D8 licensing/positioning surfaced in 00 + SPECIFICATION §9) are present as explicit decisions-with-recommendations.

Coverage gaps

The following distinctive source ideas are **not yet fully reflected** in final/. They are surfaced here per §11.4.6/§11.4.118 (no silent omission) rather than hidden. Each is tagged **deliberate-divergence** (consciously not adopted; rationale exists) or **open-gap** (should be added/recorded in a future revision).

- **G1 — [03_ZAI] Disaster-recovery completeness — partial / open-gap.** ZAI's KMS-encrypted automated backups landed (04, 02, 07) and an RTO target appears in 04-security, but ZAI's *named* DR posture — explicit **RTO/RPO budget**, Terraform-driven region-failover runbook, and "disaster recovery" as a first-class operational section — is not consolidated anywhere; the pieces are scattered. **Recommendation:** add an explicit HA/DR + RTO/RPO subsection to 08-Phase-2 (or 05-tooling) citing [03_ZAI].
- **G2 — [06_GRK] sing-box transport framework — deliberate-divergence (record the rationale).** GRK proposed **sing-box** as the universal transport-multiplexer (Hysteria2/Shadowsocks/ TUIC/etc. via one Go framework). The protocols GRK wanted are reflected (Hysteria2 → D1 Camp B; Shadowsocks + UDP-over-TCP → 08 Phase-2 transport set), but the **sing-box framework itself was not adopted** — the spec chose a custom Rust `helix-transport` crate (04_ARCH). **Rejection rationale (recorded here):** sing-box is a Go framework, which conflicts with the Rust-core decision (D2) and the iOS NE memory ceiling; a custom Rust `helix-transport` crate gives byte-for-byte client^[?]edge reuse and a single obfuscate/de-obfuscate implementation that sing-box's process model cannot match. A future pass MAY also surface this note inline in 01-data-plane; for now the considered-and-rejected record lives here.
- **G3 — [09_GCT] Fyne desktop UI — deliberate-divergence (rejected alternative not recorded).** GCT proposed **Fyne** for the desktop/web client. Flutter (04_UI) won unanimously and Fyne is correctly absent from the build plan. **Rejection rationale (recorded here):** Fyne ([09_GCT]) was evaluated and dropped in favor of Flutter — Flutter is the only toolkit that reaches all 8 required platforms (incl. iOS/Android/Aurora/HarmonyOS) with one codebase

+ a shared `helix_design` system; Fyne targets desktop and lacks the mobile/ niche-OS reach. A future pass MAY also note this inline in 03-client-core-and-ui's UI-toolkit decision.

No source is **wholly** unabsorbed: every one of the 16 ids is cited in at least 00 and the 11 deep-research appendix, and the three gaps above are scoped, named, and recommendation- tagged. The thinly-cited sources (03_ZAI, 06_GRK, 08_CPL, 09_GCT) are the "full-ecosystem / Go-or-sing-box camp" whose protocol/ops ideas were largely **generalized** into the chosen Rust-core + custom-transport design rather than carried verbatim.

Provenance of this ledger

- Synthesis evidence base: `v09-research/_SYNTHESIS.md` (digests of all 16 sources).
- Citation grep across all 13 `final/*.md` docs (source-id frequency table, 2026-06-25).
- Distinctive-term presence check (`sing-box`, `KMS`, `Fyne`, `KMP`, `RT0`, `GitOps`, `Terraform`, `Patroni`, `reproducible build`) confirming G1–G3.